CropHeal AI: A Scalable Plant Disease Detection and Natural Language Remedy Assistant

Team Name: Trio RDS Tensor **Theme:** AI in Agriculture

1. Problem Statement:

Crop diseases lead to significant agricultural losses and threaten food security. While visual identification of plant diseases is possible, farmers and home gardeners often struggle with two major issues:

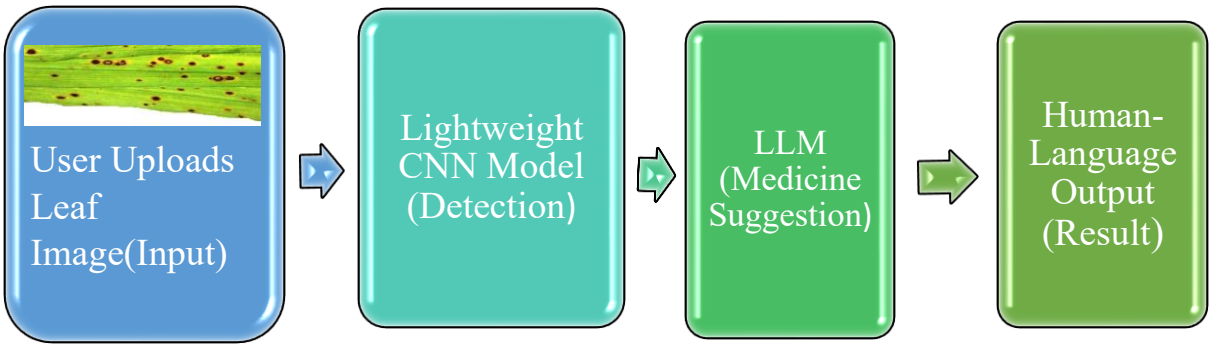
- Accurately identifying the specific disease early on.
- Understanding complex chemical terms and finding the right actionable remedy (medicine or organic solution) in simple, easy-to-understand language.

2. Proposed Solution:

We propose a two-tier Artificial Intelligence pipeline combining Computer Vision (CV) and Large Language Models (LLM) to bridge the gap between disease detection and actionable treatment.

- **Tier 1 (Detection):** A lightweight Convolutional Neural Network (CNN) trained to detect diseases from leaf images.
- **Tier 2 (Remedy Assistant):** Once the disease is classified, the system queries an LLM to generate a personalized, human-readable response suggesting appropriate chemical and organic treatments.

3. System Architecture:



4. Key Features & Unique Selling Propositions (USPs):

- **Highly Scalable Architecture:** Designed with dynamic training scripts using Transfer Learning. Adding a new plant to the system only requires uploading its image folder—the model dynamically adjusts its final layers without needing a complete retrain.
- **Human-Centric Insights:** Instead of just outputting a generic label like "Late Blight," the system provides conversational, context-aware medicine and care suggestions via LLM integration.
- **Resource Efficient:** Optimized to train and run in low-compute environments (e.g., 4GB RAM) using dataset generators and lightweight base models (like MobileNetV2).

5. Initial Scope (Prototype Crops):

To demonstrate the model's versatility across different crop types, the prototype will be trained on 8 diverse plants:

- **Vegetables/Tubers:** Potato, Tomato
- **Grains:** Rice, Corn
- **Nuts:** Peanuts

6. Technical Stack:

- **Machine Learning/CV:** Python, TensorFlow/Keras or PyTorch (Transfer Learning).
- **NLP/Generative AI:** LLM API (e.g., Gemini API) for generating natural language remedy suggestions.
- **Frontend Interface:** Flask for a rapid, interactive, and user-friendly web interface.

7. Future Impact:

Our solution aims to empower anyone—from a commercial farmer to an urban balcony gardener—with an accessible "pocket agronomist," ultimately reducing crop waste and promoting healthier plant growth.

8. References & Tools:

- **Dataset:** Original leaf image data sourced and adapted from the *PlantVillage Dataset* for training the initial classification model.
- **Computer Vision Base Model:** *MobileNetV2* architecture documentation and pre-trained weights via TensorFlow/Keras (or PyTorch).
- **Generative AI & LLM Integration:** ** DeepSeek API & Google Gemini API* (via Google AI Studio) for Natural Language Processing and generating organic/chemical remedy suggestions.
- **Development & Training Environment:** Model training and testing accelerated using *Google Colaboratory (Colab).*
- **Code Generation & UI Design Assistants:** Rapid prototyping, Vibe Coding, and UI (Flask/HTML) structuring assisted by Large Language Models including *ChatGPT, DeepSeek, and Gemini*.